Bandpass Modulation

1. Bandlimited Signals ⇒ * Coherent demodulation only
   * BPSK, QPSK, M-PSK, M-QAM, MSK
   * No Constraint on power, but Bandwidth is limited
   * ECC (Error Control coding) is needed for link performance.

2. Power limited Signals ⇒ * orthogonal signals
   * Coherent or Non coherent detection
   * No Constraint on bandwidth
   * Fixed Envelope power
   * Preferred for initial training and synchronization
   * BFSK, MFSK, CPM, ASK, MSK, DPSK

In any communication coherent as well as non-coherent detection is employed.

<u>Procedure used in Bandpass Modulation with Tx and Rx architecture</u>

1. Mathematical representation of a Signal.
2. Obtain basis functions and Express signal in terms of basis functions.
3. Draw the Constellation diagram and obtain d_{min}, E_{bavg}, E_{savg} etc.
4. Obtain PSD of a modulated Signal (Crucial for Carrier/Symbol timing recovery)

Bandpass Bandlimited Modulation

1. Binary phase Shift keying (BPSK)

Number of bits $k=1$, $M=2^k$ = Number of Symbols = 2

$$S_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) \quad ; \quad 0 \le t \le T_b, \text{ represents bit 1}$$

$$S_2(t) = -\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) \quad ; \quad 0 \le t \le T_b, \text{ represents bit 0}$$

where $E_b \rightarrow$ Energy per bit

$T_b \rightarrow$ Bit duration

$T_s \rightarrow$ Symbol duration

$E_s \rightarrow$ Energy per Symbol

$E_s = k E_b$

$T_s = k T_b$

$$\phi_1(t) = \frac{S_1(t)}{\sqrt{E_{s_1}}} \quad , \quad E_{s_1} = \int_0^{T_b} s_1^2(t)\,dt = \frac{2E_b}{T_b} \int_0^{T_b} \cos^2(2\pi f_c t)\,dt$$

$$E_{s_1} = \frac{2E_b}{T_b} \cdot \frac{T_b}{2} = E_b$$

$$\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t) \quad ; \quad 0 \le t \le T_b$$

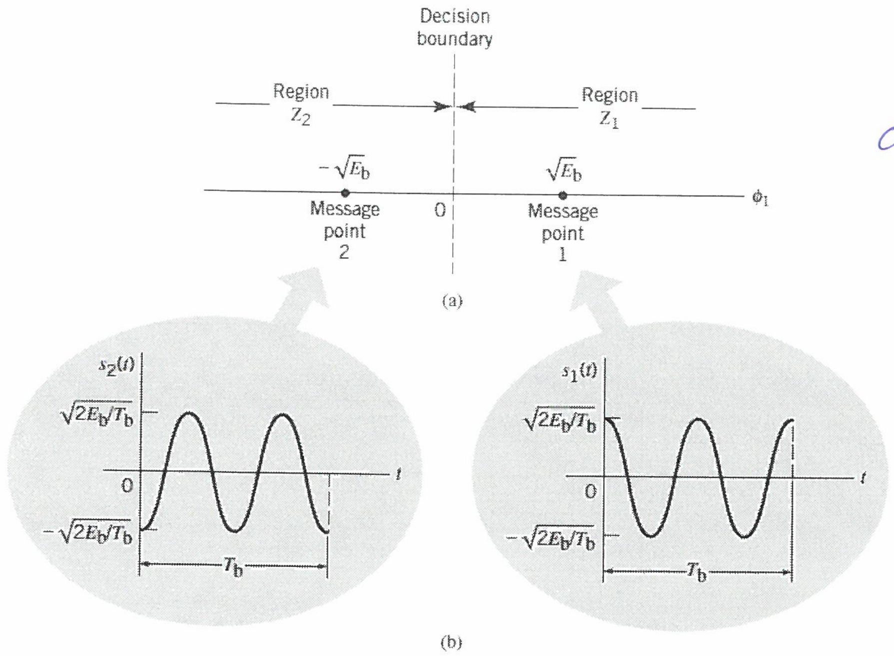
$S_2(t)$ and $S_1(t)$ are antipodal. only one basis function is sufficient to represent both the signals.

$$S_1(t) = S_{11} \cdot \phi_1(t) \quad ; \quad S_{11} = \int_0^{T_b} s_1(t) \cdot \phi_1(t) = \sqrt{E_b}$$

$$S_2(t) = S_{21} \cdot \phi_1(t) \quad ; \quad S_{21} = \int_0^{T_b} S_2(t) \cdot \phi_1(t) = -\sqrt{E_b}$$

$$S_1(t) = \sqrt{E_b}\, \phi_1(t)$$

$$S_2(t) = -\sqrt{E_b}\, \phi_1(t)$$



$d_{12} = d_{min} = 2\sqrt{E_b}$

Fig1: (a) Signal-space diagram for coherent binary PSK system. (b) The waveforms depicting the transmitted signals $s1(t)$ and $s2(t)$

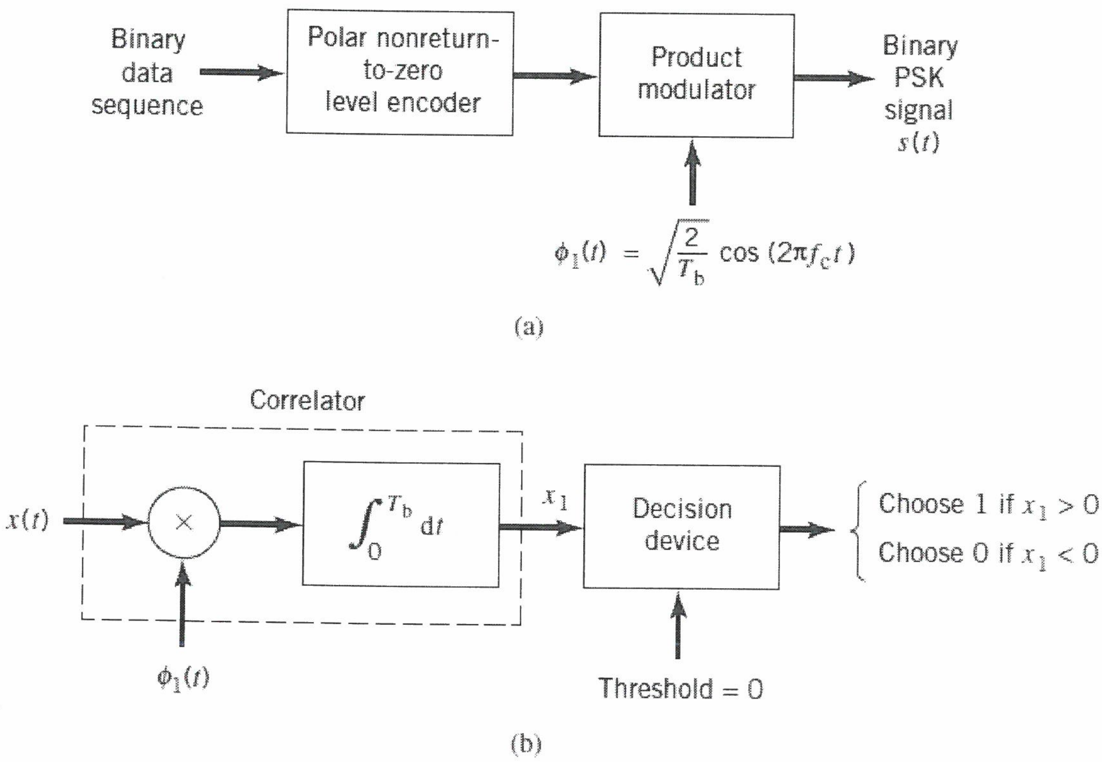


Fig2: Block diagrams for (a) binary PSK transmitter and (b) coherent binary PSK receiver.

Power Spectral density of BPSK (Baseband) :-

$$S_{XX}(f) = \frac{|G(f)|^2}{T_b} \sum_{k=-\infty}^{\infty} R(k)\, e^{jk\omega T_b}$$

Let $g(t)$ denote the pulse shaping function

$$g(t) = \sqrt{\frac{2E_b}{T_b}} \; ; \; 0 \le t < T_b$$

$$= \sqrt{\frac{2E_b}{T_b}} \; \mathrm{rect}\left(t/T_b\right)$$

$$G(f) = \sqrt{\frac{2E_b}{T_b}} \; T_b \, \mathrm{Sinc}(fT_b)$$

$$S_{XX}(f) = \frac{2E_b/T_b}{T_b} \; T_b^2 \, \mathrm{Sinc}^2(fT_b)$$

$$S_{XX}(f) = 2E_b \, \mathrm{Sinc}^2(fT_b)$$

$$S_{XX}(f) = 2E_s \, \mathrm{Sinc}^2(fT_s)$$

as $k=1$, $E_s = E_b$ and $T_s = T_b$

In the above derivation, it was assumed that the incoming signal or sequence is random, with symbols 1 and 0 being equally likely and the symbols transmitted during the different time slots being statistically independent.

Note - ① $S_{XX}(f)$ falls off as the inverse square of the frequency
② $S_{XX}(f)$ goes through zero at multiples of the bit rate. (Observe Fig 3)

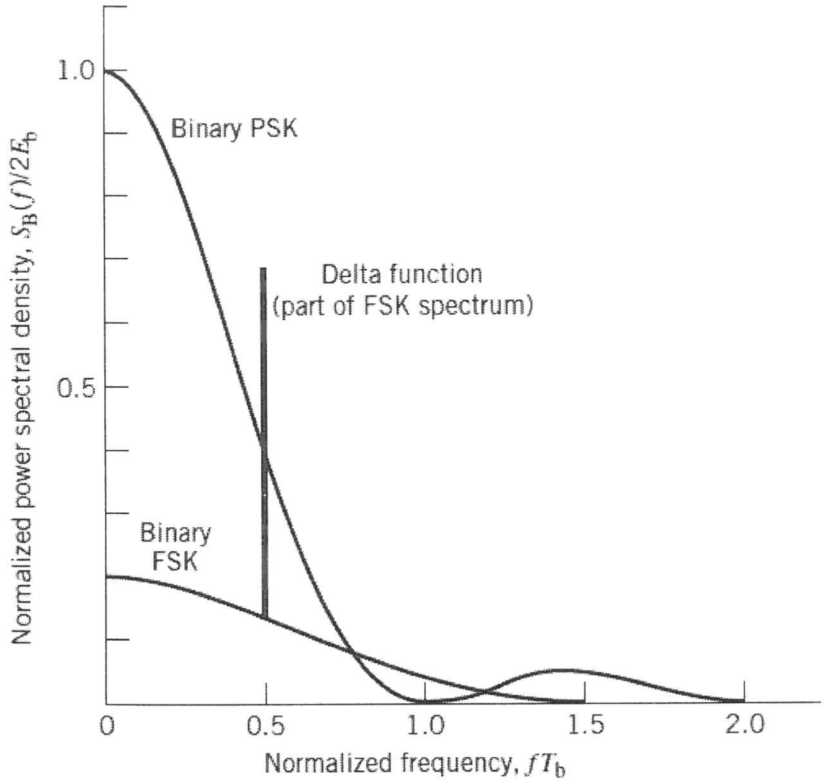


Fig3: Power spectra of binary PSK and FSK signals.

2. Quadrature phase shift keying (QPSK)

$K = 2,\ M = 4,\quad 2\ \text{bits/symbol}$

$S_1(t) \rightarrow 00 \rightarrow 45^\circ \quad (\pi/4)$

$S_2(t) \rightarrow 01 \rightarrow 135^\circ \quad (3\pi/4)$

$S_3(t) \rightarrow 11 \rightarrow 225^\circ \quad (5\pi/4)$

$S_4(t) \rightarrow 10 \rightarrow 315^\circ \quad (7\pi/4)$

$$S_i(t) = \sqrt{\frac{2E}{T}} \cos\left[2\pi f_c t + (2i-1)\frac{\pi}{4}\right] \quad ; \quad 0 \le t \le T$$

$E \rightarrow$ Transmitted energy per symbol. $i = 1, 2, 3$ and 4

$T \rightarrow$ Symbol duration.

$S_1(t)$ and $S_3(t)$ are antipodal. Similarly $S_2(t)$ and $S_4(t)$.

To represent S_1 and $S_3 \longrightarrow$ one basis function

To represent S_2 and $S_4 \rightarrow$ one basis function

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t) \quad ; \quad 0 \le t \le T$$

$$\phi_2(t) = \sqrt{\frac{2}{T}} \sin(2\pi f_c t) \quad ; \quad 0 \le t \le T$$

$$S_i(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t) \cdot \cos\left[(2i-1)\frac{\pi}{4}\right] - \sqrt{\frac{2E}{T}} \sin(2\pi f_c t) \cdot \sin\left[(2i-1)\frac{\pi}{4}\right]$$

PSD : $g(t) = \sqrt{\frac{E}{T}},\quad 0 \le t \le T$

Similar to BPSK, the PSD of QPSK can be found as

$$S_{XX}(f) = 2E_s \operatorname{Sinc}^2(f T_s) = 2(2E_b)\operatorname{Sinc}^2(f\, 2T_b)$$

$$S_{XX}(f) = 4E_b \operatorname{Sinc}^2(2f T_b)$$

$d_{min} = d_{12} = d_{14} = \sqrt{2E_s}$

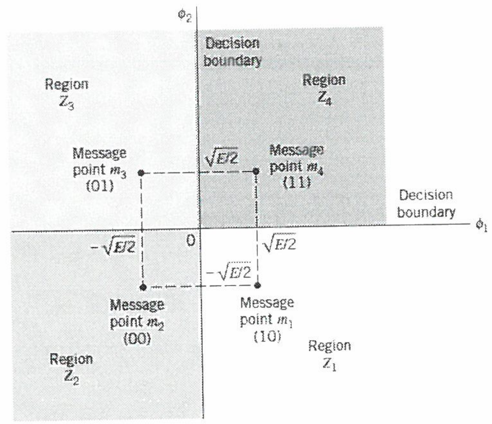


Fig4: Signal-space diagram of QPSK system.

Table: Signal-space characterization of QPSK

Gray encoded input dibit	Phase of QPSK signal (radians)	Coordinates of message points s_{i1}	s_{i2}
11	$\pi/4$	$+\sqrt{E/2}$	$+\sqrt{E/2}$
01	$3\pi/4$	$-\sqrt{E/2}$	$+\sqrt{E/2}$
00	$5\pi/4$	$-\sqrt{E/2}$	$-\sqrt{E/2}$
10	$7\pi/4$	$+\sqrt{E/2}$	$-\sqrt{E/2}$

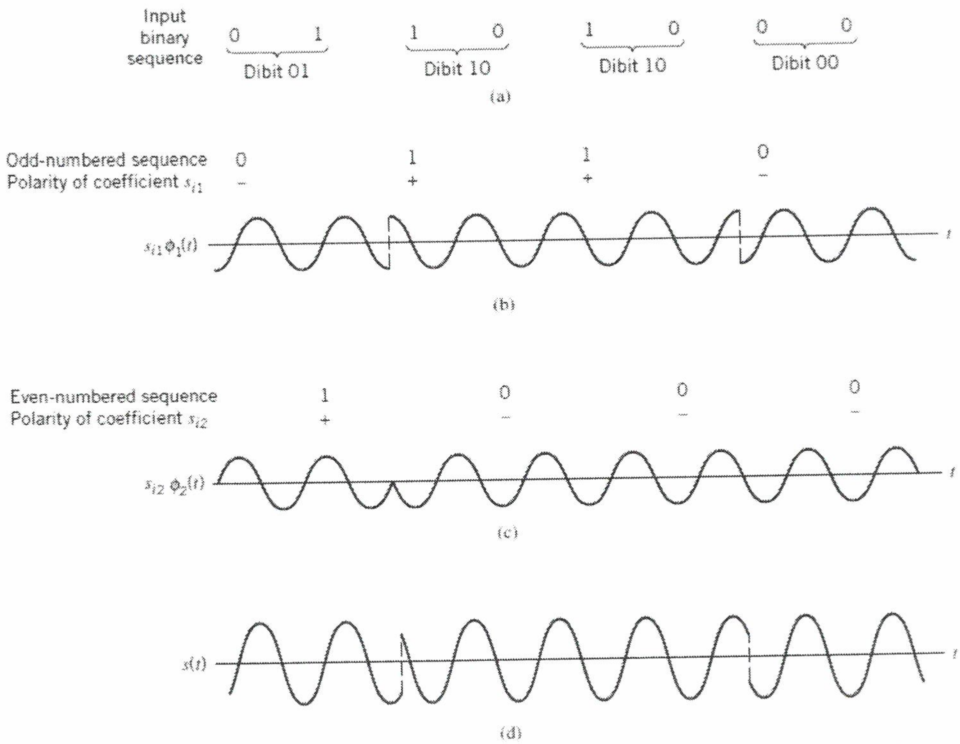
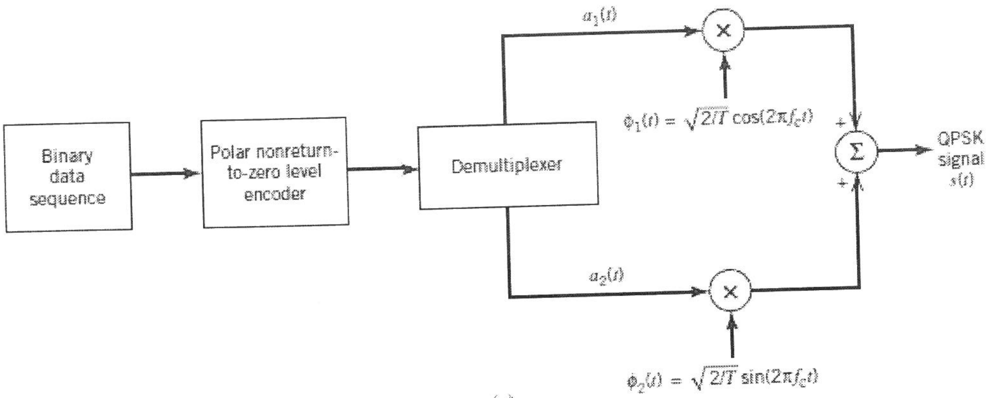
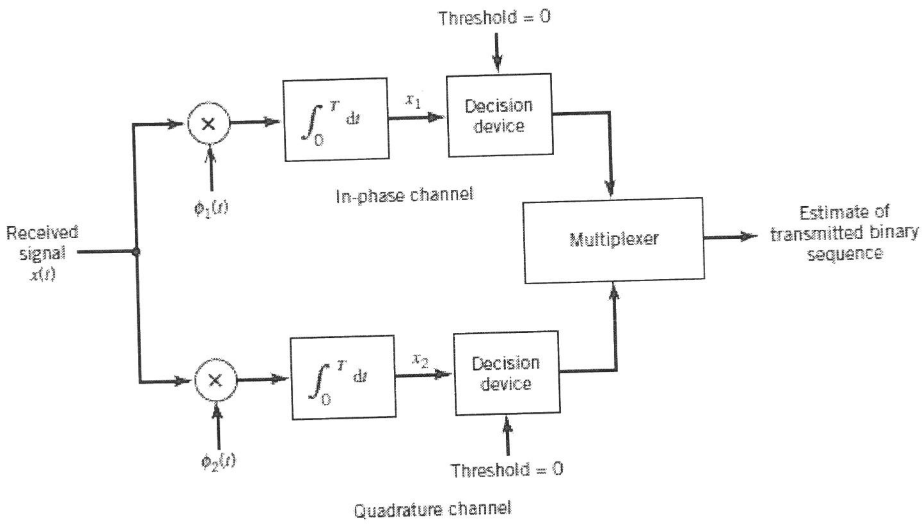


Figure 5 **(a) Input binary sequence. (b) Odd-numbered dibits of input sequence and associated binary PSK signal. (c) Even-numbered dibits of input sequence and associated binary PSK signal (d) QPSK waveform**



(a)



(b)

Fig 6: Block diagram of (a) QPSK transmitter and (b) coherent QPSK receiver.

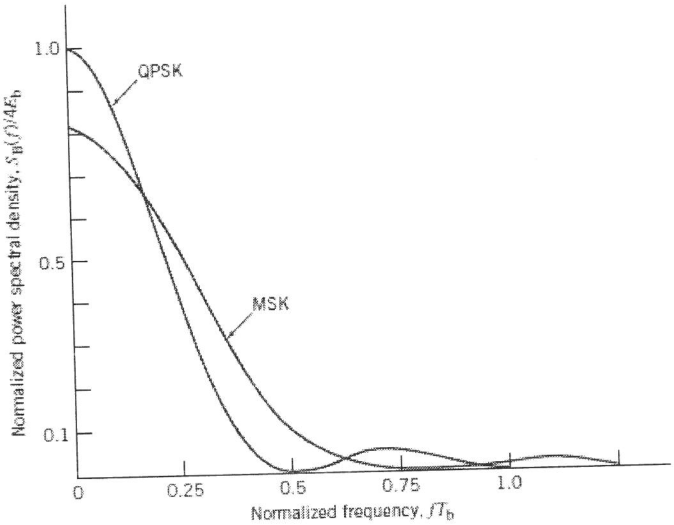


Fig 7 Power spectra of QPSK and MSK signals.

Offset QPSK (OQPSK)

Examining the QPSK waveform illustrated in Figure 4, we may make three observations:

1. The carrier phase changes by $\pm 180°$ whenever both the in-phase and quadrature components of the QPSK signal change sign. An example of this situation is illustrated in Figure 4 when the input binary sequence switches from dibit 01 to dibit 10.
2. The carrier phase changes by $\pm 90°$ whenever the in-phase or quadrature component changes sign. An example of this second situation is illustrated in Figure 4 when the input binary sequence switches from dibit 10 to dibit 00, during which the inphase component changes sign, whereas the quadrature component is unchanged.
3. The carrier phase is unchanged when neither the in-phase component nor the quadrature component changes sign. This last situation is illustrated in Figure 4 when dibit 10 is transmitted in two successive symbol intervals.

Situation 1 and, to a much lesser extent, situation 2 can be of a particular concern when the QPSK signal is filtered during the course of transmission, prior to detection. Specifically, the 180° and 90° shifts in carrier phase can result in changes in the carrier amplitude (i.e., envelope of the QPSK signal) during the course of transmission over the channel, thereby causing additional symbol errors on detection at the receiver.

To mitigate this shortcoming of QPSK, we need to reduce the extent of its amplitude fluctuations. To this end, we may use *offset QPSK*. In this variant of QPSK, the bit stream responsible for generating the quadrature component is delayed (i.e., offset) by half a symbol interval with respect to the bit stream responsible for generating the in-phase component. Specifically, the two basis functions of offset QPSK are defined by

$$\phi_1(t) = \sqrt{\frac{2}{T}}\cos(2\pi f_c t), \qquad 0 \le t \le T$$

$$\phi_2(t) = \sqrt{\frac{2}{T}}\sin(2\pi f_c t), \qquad \frac{T}{2} \le t \le \frac{3T}{2}$$

Since, in addition to $\pm 90°$ phase transitions, $\pm 180°$ phase transitions also occur in QPSK, we find that amplitude fluctuations in offset QPSK due to filtering have a smaller amplitude than in the case of QPSK.

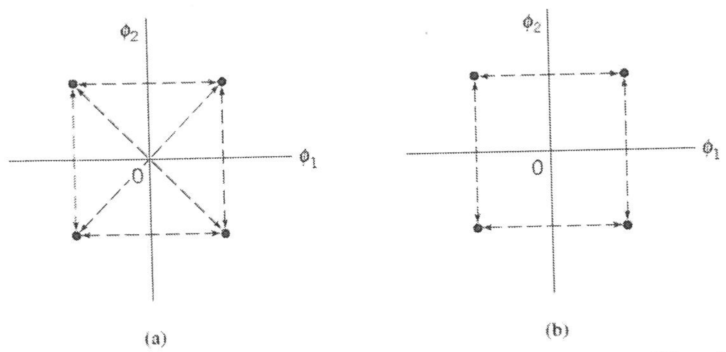


Figure 8: Possible paths for switching between the message points in (a) QPSK and (b) offset QPSK

3. Octo PSK (8-PSK)

$K=3,\ M=2^3=8$

$$s_i(t)=\sqrt{\frac{2E}{T}}\cos\left[2\pi f_c t+\frac{2\pi}{M}(i-1)\right],\quad i=1,2\ldots M$$

When $M=8$,

$$s_i(t)=\sqrt{\frac{2E}{T}}\cos\left[2\pi f_c t+\frac{\pi}{4}(i-1)\right],\quad i=1,2\ldots 8$$

$$S_1(t)=\sqrt{\frac{2E}{T}}\cos\left[2\pi f_c t\right]$$

$$S_2(t)=\sqrt{\frac{2E}{T}}\cos\left[2\pi f_c t+\pi/4\right]$$

$$S_3(t)=\sqrt{\frac{2E}{T}}\sin\left[2\pi f_c t\right]$$

$$S_4(t)=\sqrt{\frac{2E}{T}}\cos\left[2\pi f_c t+3\pi/4\right]$$

$S_5(t)=-S_1(t),\quad S_6(t)=-S_2(t),\quad S_7(t)=-S_3(t),\quad S_8(t)=-S_4(t)$

$$\phi_1(t)=\sqrt{\frac{2}{T}}\cos(2\pi f_c t)\ ;\quad 0\le t\le T$$

$$\phi_2(t)=\sqrt{\frac{2}{T}}\sin(2\pi f_c t)\ ;\quad 0\le t\le T$$

Two basis functions are sufficient to represent all 8 signals.

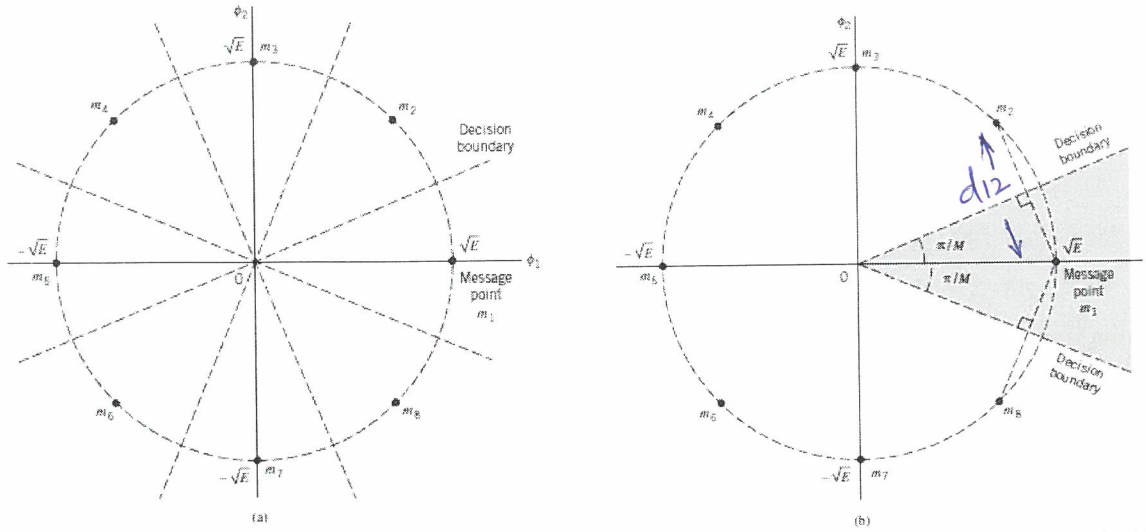
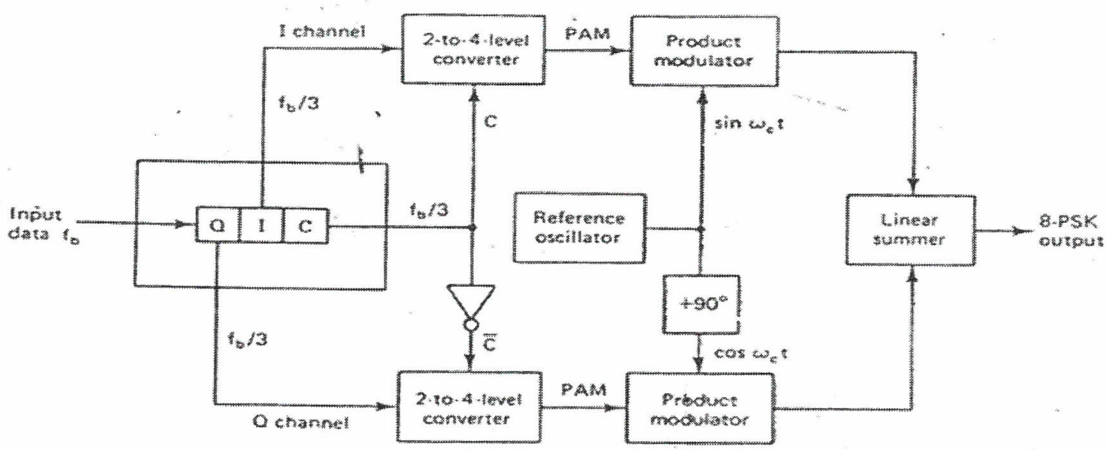
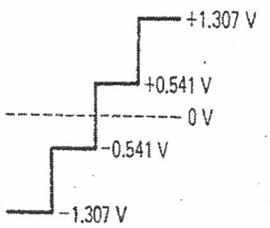


Fig 9 (a) Signal-space diagram for octaphase-shift keying (i.e., $M = 8$). The decision boundaries are shown as dashed lines.
(b) Signal-space diagram illustrating the application of the union bound for octaphase-shift keying.



I	C	Output
0	0	−0.541 V
0	1	−1.307 V
1	0	+0.541 V
1	1	+1.307 V

Q	$\bar{C}$	Output
0	1	−1.307 V
0	0	−0.541 V
1	1	+1.307 V
1	0	+0.541 V



Binary input			8PSK output phase
Q	I	C	
0	0	0	−112.5°
0	0	1	−157.5°
0	1	0	−67.5°
0	1	1	−22.5°
1	0	0	+112.5°
1	0	1	+157.5°
1	1	0	+67.5°
1	1	1	+22.5°

wrto Fig 9a, $\sin(\pi/M) = \dfrac{d_{12}/2}{\sqrt{E}}$

$$d_{12} = d_{18} = 2\sqrt{E}\,\sin(\pi/M) = d_{min} = 2\sqrt{E_{sav}}\,\sin(\pi/M)$$

$$= 2\sqrt{E_{bavg}.\log_2 M.\sin^2(\pi/M)}$$

$$\approx 2\sqrt{E_{bavg}.\log_2 M.(\pi/M)^2}$$

- Example 13-5:

For a tribit input of Q = 0, I = 0, and C = 0(000), determine the output phase for the 8-PSK modulator shown in Figure 13-20.

Solution:

The inputs to the I-channel 2-to-4-level converter are I = 0 and C = 0. From Figure 13-21 the output is –0.541V. The inputs to the Q-channel 2-to-4-level converter are Q = 0 and C = 1. Again from Figure 13-21, the output is –1.307V.

Thus the two inputs to the I-channel product modulators are –0.541V and $\sin\omega_c t$. The output is

$$I = (-0.541)(\sin\omega_c t) = -0.541 \sin\omega_c t.$$

- The two inputs to the Q-channel product modulator are –1.307V and $\cos\omega_c t$. The output is

$$Q = (-1.307)(\cos\omega_c t) = -1.307 \cos\omega_c t.$$

The outputs of the I- and Q-channel product modulators are combined in the linear summer and produce a modulated output of

$$\text{summer output} = -0.541 \sin\omega_c t - 1.307 \cos\omega_c t$$
$$= 1.41 \sin(\omega_c t - 112.5^\circ)$$

For the remaining tribit codes (001,010,011,100,101,110,and 111), the procedure is the same.

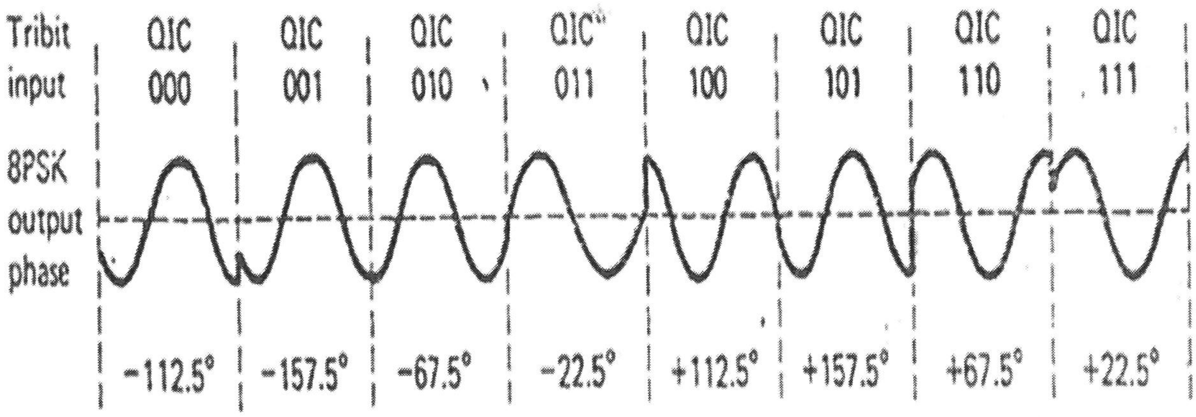
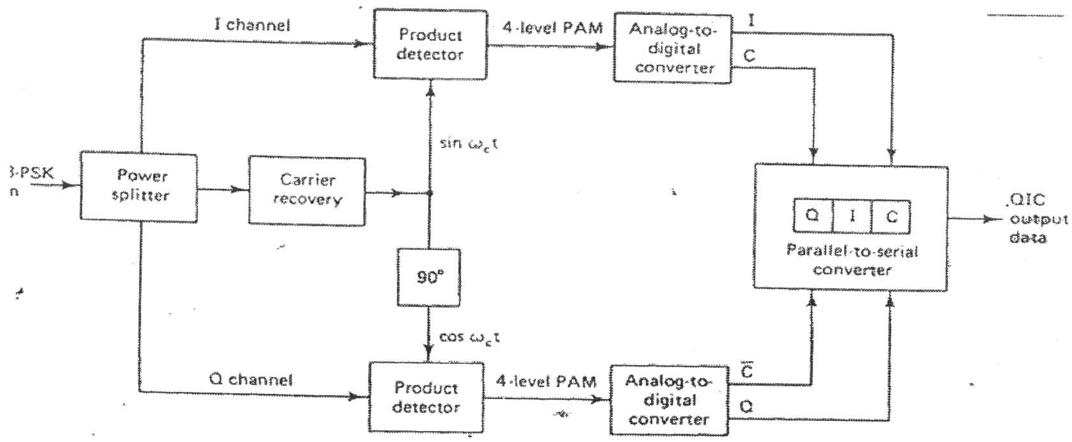


Fig. 13-23. Output phase versus time for an 8-PSK modulator.



- Figure 13-25 shows a block diagram of an 8-PSK receiver. The power splitter directs the input 8-PSK signal to the I and Q product detectors and the carrier recovery circuit.
- The carrier recovery circuit reproduces the original reference oscillator signal.
- The incoming 8-PSK signal is mixed with the recovered carrier in the I product detector and with a quadrature carrier in the Q product detector.
- The outputs of the product detectors are 4-level PAM signals that are fed to the 4-to-2- level analog-to-digital converters (ADCs).
- The outputs from the I-channel 4-to-2- level converter are the I and C bits, while the outputs from the Q-channel 4-to-2-level converter are the Q and $\overline{C}$ bits. The parallel-to-serial logic circuit converts the I/C and Q/$\overline{C}$ bit pairs to serial I, Q, and C output data streams.
- With 8-PSK, since the data are divided into three channels, the bit rate in the 1, Q, or C channel is equal to one-third of the binary input data rate ($f_b/3$). (The bit splitter stretches the I, Q, and C bits to three times their input bit length.)
- Because the I, Q, and C bits are outputted simultaneously and in parallel the 2-to-4-level converters also see a chance in their inputs (and consequently their outputs) at a rate equal to $f_b/3$.
- Figure 13-24 shows the bit timing, relationship between the binary input data; the I-, Q-, and C-channel data; and the I and Q PAM signals. It can be seen that the highest fundamental frequency in the I, Q or C channel is equal to one-sixth of the bit rate of the binary input (one cycle in the I, Q, or C channel takes the same amount of time as six input bits).
- Also, the highest fundamental frequency in either PAM signal is equal to one-sixth of the binary input bit rate

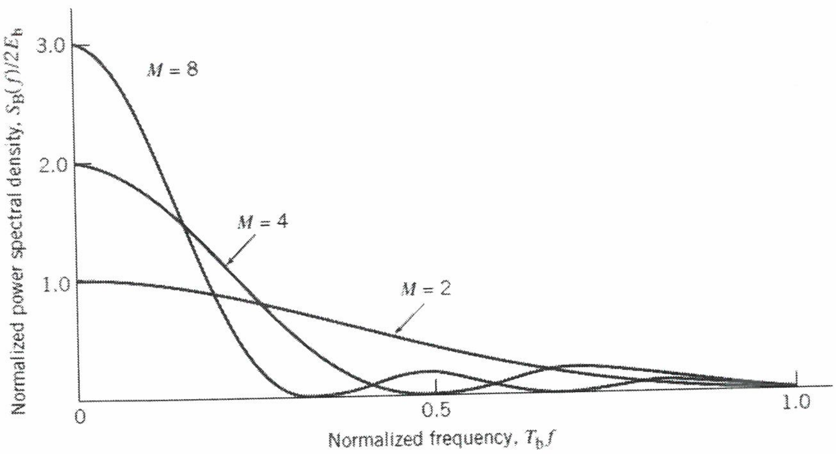


Fig: Power spectra of *M*-ary PSK signals for M = 2, 4, 8.

For M-psk,

$$S_{xx}(f) = 2 \cdot k E_b \, \text{Sinc}^2 (f k T_b)$$

$$= 2 E_b \cdot \log_2 M \cdot \text{Sinc}^2 (f T_b \log_2 M)$$

For 8-psk,

$$S_{xx}(f) = 6 E_b \, \text{Sinc}^2 (3 f T_b)$$

For Qpsk,

$$S_{xx}(f) = 4 E_b \, \text{Sinc}^2 (2 f T_b)$$

For Bpsk,

$$S_{xx}(f) = 2 E_b \, \text{Sinc}^2 (f T_b)$$

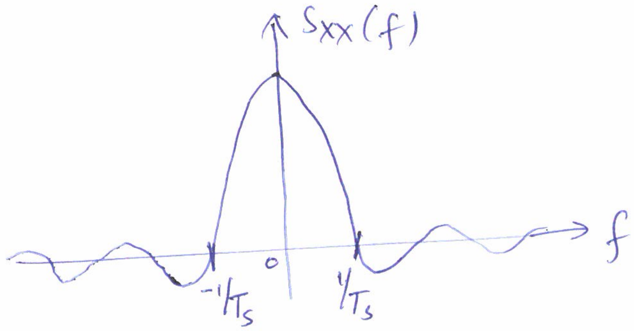
MPSK

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos\left[2\pi f_c t + \frac{2\pi}{M}(i-1)\right], \quad i = 1,2,3 \ldots M$$

2-basis functions $\phi_1(t)$ and $\phi_2(t)$

$$d_{min} = 2\sqrt{E_{bavg} \cdot \log_2 M \cdot (\pi/M)^2}$$

$$PSD = S_{xx}(f) = 2 \cdot E_b \log_2 M \cdot \mathrm{Sinc}^2(f T_b \log_2 M)$$



Null to null bandwidth $= B = \frac{2}{T_s} = 2R_s = \frac{2R_b}{K} = \frac{2R_b}{\log_2 M}$

$R_b = \frac{B \log_2 M}{2}$, $R_b \rightarrow$ bit rate

$R_s \rightarrow$ Symbol rate

Spectral efficiency $\rho = \frac{R_b}{B} = \frac{B \log_2 M}{2B} = \frac{\log_2 M}{2}$ b/s/Hz

M	2	4	8	16	32	64
ρ	0.5	1	1.5	2	2.5	3

As $K\uparrow$ $M\uparrow$ $\rho\uparrow$ Data rate $\uparrow$, $P_e\uparrow$ QoS $\downarrow$

** In order to maintain QoS, Increase SNR as MPSK is Bandlimited

Pulse Amplitude Modulation (PAM)

1. Baseband PAM:

In Digital PAM Signal waveform is represented by

$$S_m(t) = A_m\, p(t), \qquad 1 \le m \le M$$

$A_m \rightarrow$ Amplitude, $p(t) \rightarrow$ pulse Shaping function of duration T.

Signal amplitudes A_m take discrete values

$$A_m = 2m - 1 - M, \quad m = 1, 2 \ldots M \rightarrow ①$$

The amplitudes are $\pm 1, \pm 3, \pm 5 \ldots \pm (M-1)$

Eqn ① has Symmetric constellation and gives less average Energy.

Signal Energy $E_m = \int_{-\infty}^{\infty} A_m^2\, p^2(t)\, dt = A_m^2 E_p$

$E_p \rightarrow$ Energy in $p(t)$

$$E_{savg} = p(s_i) \sum_{m=1}^{M} S_m^2 = \frac{E_p}{M} \sum_{m=1}^{M} A_m^2$$

$$= \frac{2E_p}{M} \left(1^2 + 3^2 + 5^2 + \cdots + (M-1)^2\right)$$

$$E_{savg} = \frac{2E_p}{M} \times \frac{M(M^2-1)}{6} = \frac{(M^2-1)E_p}{3}$$

$$E_{bavg} = \frac{E_{savg}}{\log_2 M} = \frac{(M^2-1)\, E_p}{3 \log_2 M}$$

Basis function $\phi(t) = \frac{p(t)}{\sqrt{E_p}}$

$S_m(t) = A_m \sqrt{E_p}\, \phi(t)$, $\quad d_{mn} = \sqrt{\| S_m - S_n \|^2} = |A_m - A_n| \sqrt{E_p}$

2. Bandpass PAM (ASK - Amplitude Shift Keying)

PAM signals are carrier modulated bandpass signals with lowpass equivalents of the form $A_m g(t)$, where A_m and $g(t)$ are real.

$$s_m(t) = Re\left[s_{ml}(t)\, e^{j2\pi f_c t} \right] = Re\left[A_m g(t)\, e^{j2\pi f_c t} \right]$$

$$= A_m g(t) \cos(2\pi f_c t) = A_m p(t)$$

$$E_m = \frac{A_m^2}{2} E_g \quad ; \quad \text{Energy in bandpass} = \frac{\text{Energy in lowpass}}{2}$$

$$E_{s\,avg} = \frac{(M^2-1)\, E_g}{6}$$

$$E_{b\,avg} = \frac{(M^2-1)\, E_g}{6 \log_2 M} \rightarrow (2)$$

PAM is one-dimensional, they need only one basis function.

$$\phi(t) = \sqrt{\frac{2}{E_g}}\, g(t) \cos(2\pi f_c t)$$

$$s_m(t) = A_m \sqrt{\frac{E_g}{2}}\, \phi(t)$$

$$d_{mn} = |A_m - A_n| \sqrt{\frac{E_g}{2}}$$

For adjacent signal points $|A_m - A_n| = 2$, hence the minimum distance of the constellation is given by

$$d_{min} = 2\sqrt{E_p} = \sqrt{2 E_g}$$

From (2), $$d_{min} = \sqrt{\frac{12 \log_2 M}{(M^2-1)} \cdot E_{b\,avg}}$$

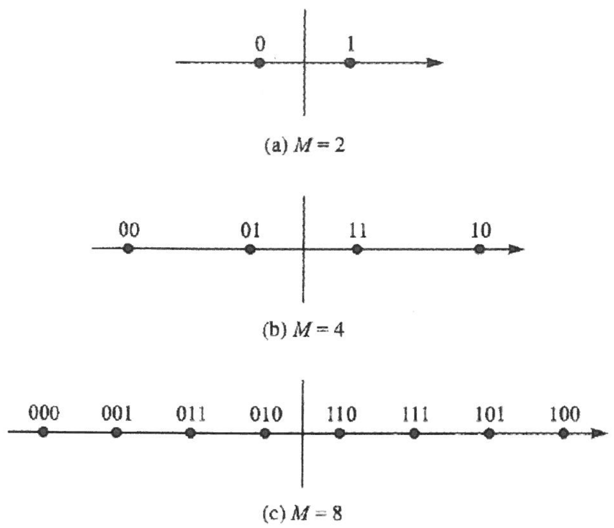


FIGURE 3.2–1
Constellation for PAM signaling.

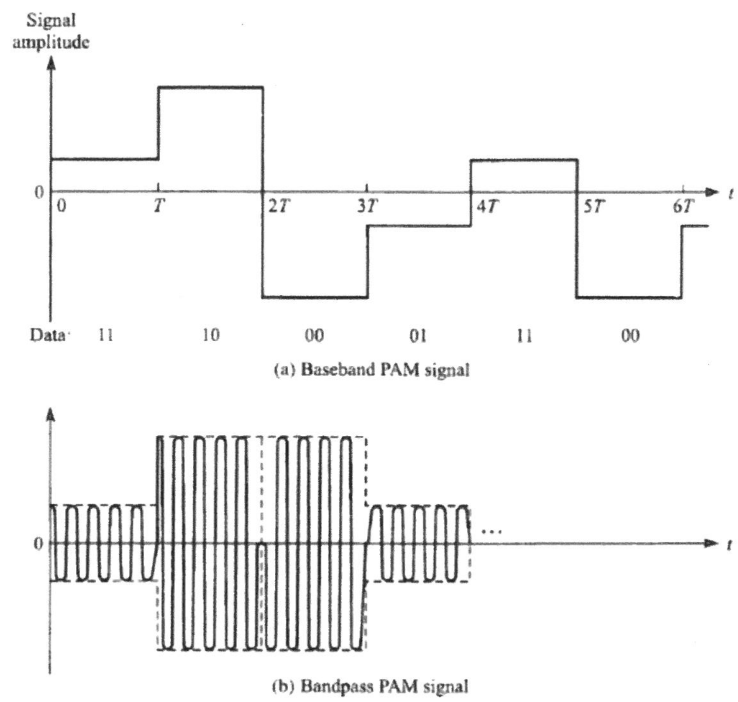


FIGURE 3.2–2
Example of (a) baseband and (b) carrier-modulated PAM signals.

QAM – Quadrature Amplitude Modulation

The bandwidth Efficiency of PAM can also be obtained by Simultaneously impressing two seperate k-bit symbols from the Information sequence on two quadrature carriers $\cos(2\pi f_c t)$ and $\sin(2\pi f_c t)$. The resulting modulation is called Quadrature PAM or QAM.

Signal waveform is

$$S_m(t) = Re\left[(A_{mi} + j A_{mq})\, g(t)\, e^{j2\pi f_c t}\right]$$

$$= A_{mi}\, g(t) \cos(2\pi f_c t) - A_{mq}\, g(t) \sin(2\pi f_c t), \quad m = 1, 2 \ldots M$$

where A_{mi} and A_{mq} are the Information bearing signal amplitudes of the quadrature carriers and $g(t)$ is the signal pulse.

$$S_m(t) = Re\left[r_m e^{j\theta_m} e^{j2\pi f_c t}\right] = r_m \cos(2\pi f_c t + \theta_m)$$

$$r_m = \sqrt{A_{mi}^2 + A_{mq}^2}, \quad \theta_m = \tan^{-1}(A_{mq}/A_{mi})$$

From the above Expressions, it is appearent that QAM signal waveforms may be viewed as Combined amplitude (r_m) and phase (θ_m) modulation.

M_1 PAM $\times$ M_2 PSK $= M = M_1 M_2$ Combined PAM-PSK Constellation.

If $M_1 = 2^n$, $M_2 = 2^m$, the Combined Constellation results in Simultaneous transmission of $m + n = \log_2 M_1 M_2$ binary digits.

The symbol rate R_s becomes $\frac{R_s}{m+n}$

The basis functions are

$$\phi_1(t) = \sqrt{\frac{2}{E_g}}\, g(t) \cos(2\pi f_c t)$$

$$\phi_2(t) = \sqrt{\frac{2}{E_g}}\, g(t) \sin(2\pi f_c t)$$

$$S_m(t) = A_{mi}\sqrt{\frac{E_g}{2}}\,\phi_1(t) + A_{mq}\sqrt{\frac{E_g}{2}}\,\phi_2(t)$$

In vector representation $S_m = (S_{m1}, S_{m2})$

$$= \left(A_{mi}\sqrt{\frac{E_g}{2}},\ A_{mq}\sqrt{\frac{E_g}{2}}\right)$$

$$E_m = \|S_m\|^2 = \frac{E_g}{2}\left(A_{mi}^2 + A_{mq}^2\right)$$

The Euclidean distance b/w any pair of signal vectors in QAM is

$$d_{mn} = \sqrt{\|S_m - S_n\|^2}$$

$$= \sqrt{\frac{E_g}{2}\left[(A_{mi} - A_{ni})^2 + (A_{mq} - A_{nq})^2\right]}$$

In the special case where the signal amplitudes take the set of discrete values $\{(2m-1-M)\}$, $m = 1, 2 \ldots M\}$, the signal space diagram is rectangular. In this case, the Euclidean distance b/w adjacent points, the minimum distance is

$$d_{min} = \sqrt{2E_g} \qquad \text{Same as PAM.}$$

In the rectangular constellation with $M = 2^{2k_1}$, ie $M = 4, 16, 64, 256$ --- and with amplitudes of $\pm 1, \pm 3 \ldots$, to $\pm(\sqrt{M}-1)$ on both directions, we have

$$E_{savg} = \frac{1}{M}\frac{E_g}{2}\sum_{m=1}^{\sqrt{M}}\sum_{n=1}^{\sqrt{M}}\left(A_m^2 + A_n^2\right)$$

$$E_{savg} = \frac{E_g}{2M} \times \frac{2M(M-1)}{3} = \frac{M-1}{3}E_g$$

$$E_{bavg} = \frac{M-1}{3\log_2 M}E_g$$

$$d_{min} = \sqrt{\frac{6\log_2 M}{M-1}E_{bavg}}$$

The modulator consists of a vector mapper, which maps each of the M messages onto a constellation of size M, followed by a two-dimensional vector to signal mapper.

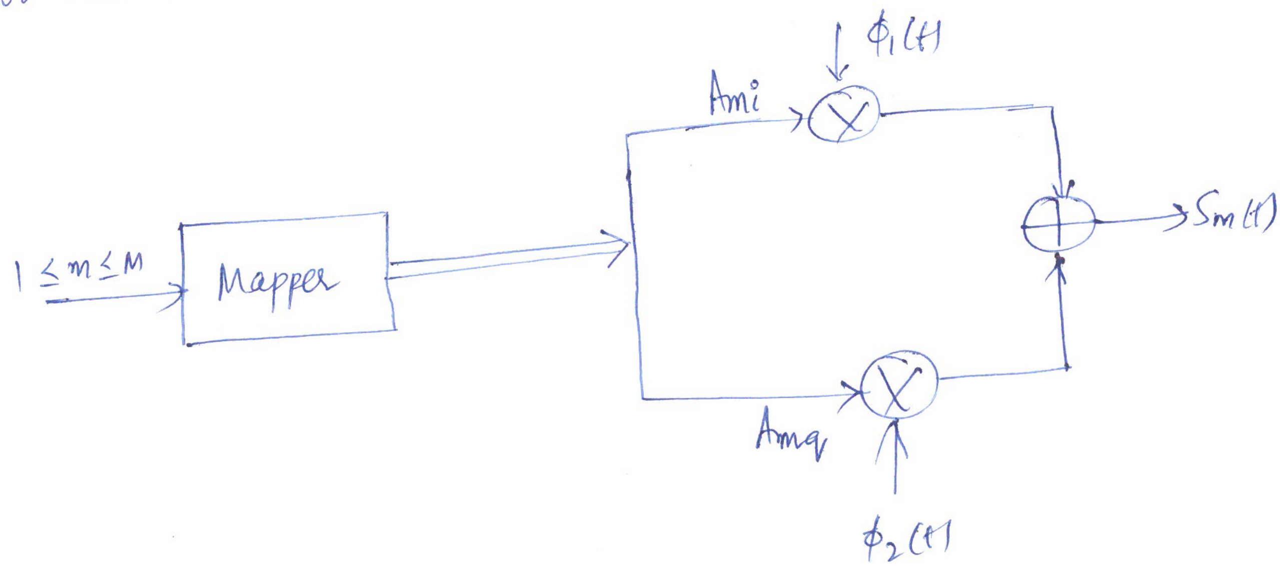


Fig - General QAM modulator

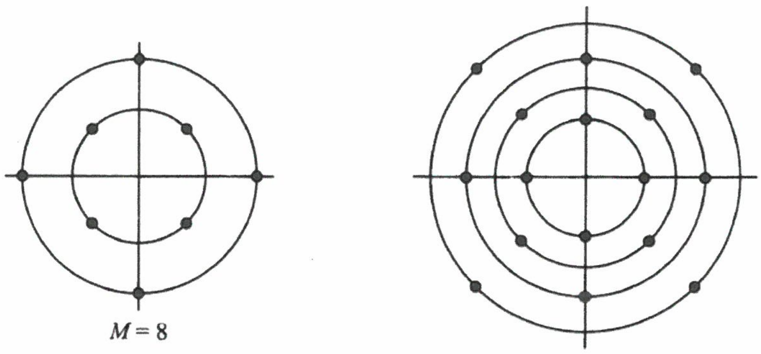


FIGURE 3.2–4
Examples of combined PAM-PSK constellations.

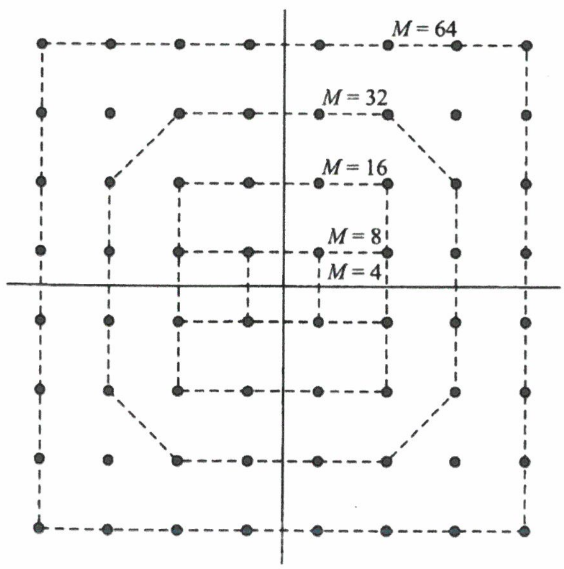


FIGURE 3.2–5
Several signal space diagrams for rectangular QAM.

TABLE 3.2–1
Comparison of PAM, PSK, and QAM

Signaling Scheme	$s_m(t)$	s_m	E_{avg}	E_{bavg}	d_{min}
Baseband PAM	$A_m p(t)$	$A_m\sqrt{\mathcal{E}_p}$	$\frac{2(M^2-1)}{3}\mathcal{E}_p$	$\frac{2(M^2-1)}{3\log_2 M}\mathcal{E}_p$	$\sqrt{\frac{6\log_2 M}{M^2-1}\mathcal{E}_{bavg}}$
Bandpass PAM	$A_m g(t)\cos 2\pi f_c t$	$A_m\sqrt{\frac{\mathcal{E}_g}{2}}$	$\frac{M^2-1}{3}\mathcal{E}_g$	$\frac{M^2-1}{3\log_2 M}\mathcal{E}_g$	$\sqrt{\frac{6\log_2 M}{M^2-1}\mathcal{E}_{bavg}}$
PSK	$g(t)\cos\left[2\pi f_c t + \frac{2\pi}{M}(m-1)\right]$	$\sqrt{\frac{\mathcal{E}_g}{2}}\left(\cos\frac{2\pi}{M}(m-1), \sin\frac{2\pi}{M}(m-1)\right)$	$\frac{1}{2}\mathcal{E}_g$	$\frac{1}{2\log_2 M}\mathcal{E}_g$	$2\sqrt{\log_2 M\ \sin^2\left(\frac{\pi}{M}\right)\mathcal{E}_{bavg}}$
QAM	$A_{mi}g(t)\cos 2\pi f_c t - A_{mq}g(t)\sin 2\pi f_c t$	$\sqrt{\frac{\mathcal{E}_g}{2}}(A_{mi}, A_{mq})$	$\frac{M-1}{3}\mathcal{E}_g$	$\frac{M-1}{3\log_2 M}\mathcal{E}_g$	$\sqrt{\frac{6\log_2 M}{M-1}\mathcal{E}_{bavg}}$

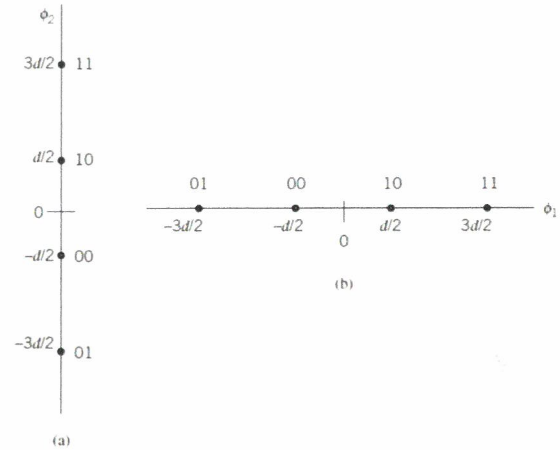


Figure 7.23
The two orthogonal constellations of the 4-ary PAM. (a) Vertically oriented constellation. (b) Horizontally oriented constellation. As mentioned in the text, we move top-down along the ϕ_2-axis and from left to right along the ϕ_1-axis.

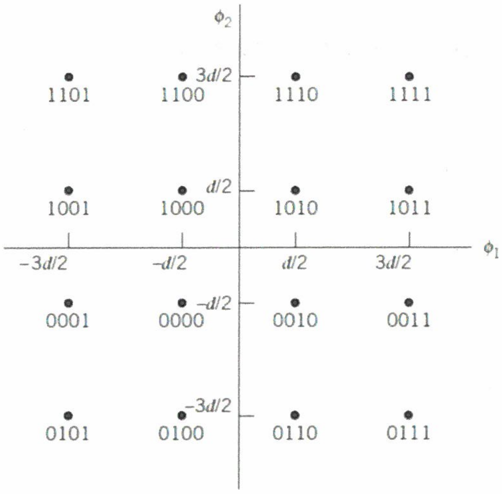


Figure 7.24
(a) Signal-space diagram of M-ary QAM for $M = 16$; the message points in each quadrant are identified with Gray-encoded quadbits.

8-QAM Transmitter and Receiver Architecture

- **Figure 13-27 shows the block diagram of an 8-QAM transmitter. As you can see, the only difference between the 8-QAM transmitter and the 8-PSK transmitter shown in Figure 13-19 is the omission of the inverter between the C channel and the Q product modulator.**

- **As with 8-PSK, the incoming data are divided into groups of three (tribits): the I, Q, and C channels, each with a bit rate equal to one-third of the incoming data rate.**

- **Again, the I and Q bits determine the polarity of the PAM signal at the output of the 2-to-4-level converters, and the C channel determines the magnitude.**

- Because the C bit is fed uninverted to both the I- and Q-channel 2-to-4-level converters, the magnitudes of I and Q PAM signals are always equal.

- Their polarities depend on the logic condition of the I and Q bits and therefore may be different. Figure 13-29 shows the truth table for the I and Q-channel 2-to-4-level converters; they are the same.

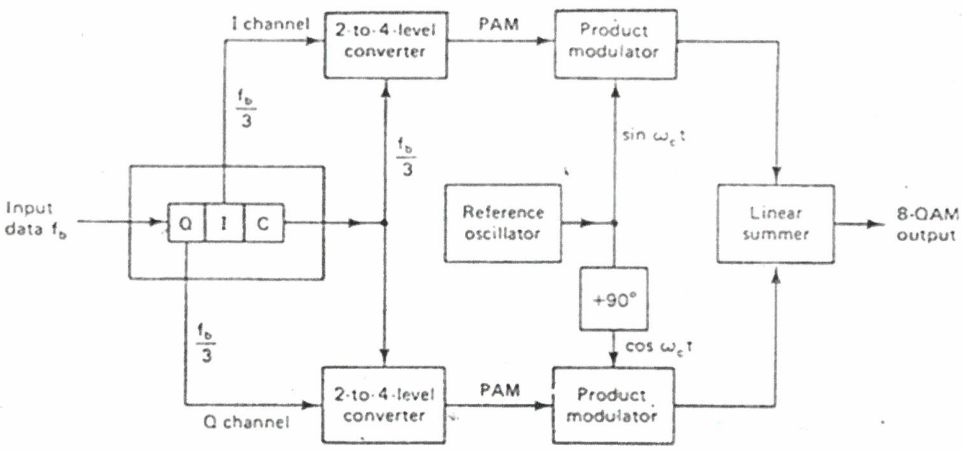


Figure 13-27. 8-QAM transmitter block diagram.

I/Q	C	Output
0	0	−0.541
0	1	−1.307 V
1	0	+0.541
1	1	+1.307 V

Fig. 13-28. Truth table for the I- and Q-channel 2-to-4-level converters.

Binary input Q I C	8QAM output amplitude	phase
0 0 0	0.765 V	−135°
0 0 1	1.848 V	−135°
0 1 0	0.765 V	−45°
0 1 1	1.848 V	−45°
1 0 0	0.765 V	+135°
1 0 1	1.848 V	+135°
1 1 0	0.765 V	+45°
1 1 1	1.848 V	+45°

Example 13-7:

For a tribit input of Q = 0, I = 0, and C = 0 (000), determine output amplitude and phase for the 8-QAM modulator shown in Figure 13-27.

Solution:

The inputs to the I-channel 2-to-4-level converter are I = 0 and C = 0. From Figure 13-28 the output is -0.541V. The inputs to the Q-channel 2-to-4-level converter are Q = 0 and C = 0. Again from Figure 13-28 the output is -0.541V.

Thus the two inputs to the I-channel product modulator are -0.541 and $\sin\omega_c t$. The output is

$$I = (-0.541)(\sin\omega_c t) = -0.541 \sin\omega_c t$$

- The two inputs to the Q-channel pro duct modulator are -0.541 and $\cos\omega_c t$. The output is

$$Q = (-0.541)(\cos\omega_c t) = -0.541 \cos\omega_c t$$

The outputs from the I- land Q-channel product modulators are combined in the linear summer and produce a modulated output of

$$\text{summer output} = -0.541 \sin\omega_c t - 0.541 \cos\omega_c t$$
$$= 0.765 \sin(\omega_c t - 135^\circ)$$

For the remaining tribit codes (001, 010, 011, 100, 101, 110, and 111), the procedure is the same.

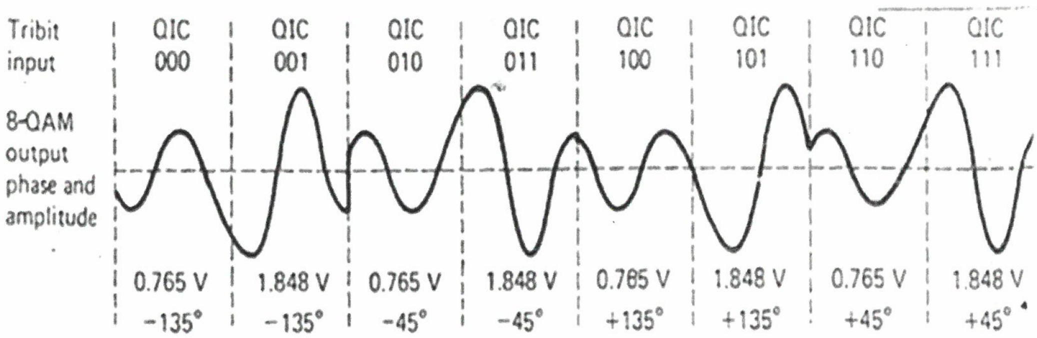


Figure 13-30. Output phase and amplitude versus time relationship for 8-QAM.

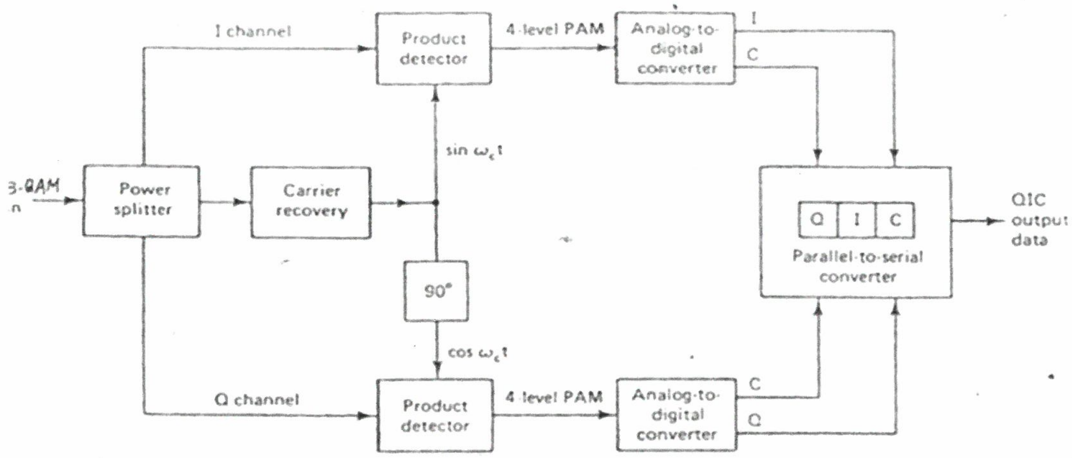


Fig. 13-31. 8-QAM receiver.

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Summary of Bandlimited Modulation Schemes (PAM, PSK, QAM)

From Bandpass PAM, PSK and QAM discussions it is clear that, all these are schemes with a general form,

$$S_m(t) = \mathrm{Re}\left[A_m\, g(t)\, e^{j2\pi f_c t}\right], \quad m = 1, 2 \ldots M$$

where A_m is determined by the signalling scheme.

M-PAM $\rightarrow$ A_m is real, $A_m = \pm 1, \pm 3, \ldots \pm (M-1)$

M-PSK $\rightarrow$ A_m is Complex, $A_m = e^{j\frac{2\pi}{M}(m-1)}$

M-QAM $\rightarrow$ A_m is a general Complex number, $A_m = A_{mi} + jA_{mq}$

In QAM, amplitude and phase, both carry Information. The dimensionality of PAM = 1, and for PSK and QAM it is 2, and it is Independent of M.

Binary Frequency Shift Keying (BFSK)

BFSK is a non-linear modulation scheme, where amplitude of the modulated signal does not vary linearly with the message signal. These non-linear modulation schemes are power efficient but bandwidth inefficient. Ex - FSK, MSK, GMSK, Constant Envelope modulation.

In BFSK, $S_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_i t) \;;\; 0 \le t \le T_b$

where $i = 1$ and 2. E_b - Transmitted signal energy per bit

$f_i = \frac{n_c + i}{T_b}$ for some fixed integer n_c and $i = 1, 2$

Symbol 1 $\rightarrow S_1(t)$, Symbol 0 $\rightarrow S_2(t)$.

The FSK discussed here is continuous-phase signal, phase continuity is always maintained, including the inter-bit switching times.

$$\phi_i(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_i t) \;;\; 0 \le t \le T_b$$

The coefficients $S_{ij} = \int_0^{T_b} S_i(t)\, \phi_j(t)\, dt$

$$S_{ij} = \int_0^{T_b} \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_i t) \sqrt{\frac{2}{T_b}} \cos(2\pi f_j t)\, dt$$

$$S_{ij} = \begin{cases} \sqrt{E_b} & i = j \\ 0 & i \ne j \end{cases}$$

Binary FSK has signal space diagram that is two dimensional with two message points.

The two message points are defined by the vectors

$$s_1 = \begin{bmatrix} \sqrt{E_b} \\ 0 \end{bmatrix} \quad \text{and} \quad s_2 = \begin{bmatrix} 0 \\ \sqrt{E_b} \end{bmatrix}$$

$$d_{min} = \sqrt{2E_b}$$

PSD of BFSK

The BFSK signal is expressed as,

$$s(t) = \sqrt{\frac{2E_b}{T_b}} \cos\left(2\pi f_c t \pm \frac{\pi t}{T_b}\right) \quad , \quad 0 \le t \le T_b$$

$$= \sqrt{\frac{2E_b}{T_b}} \cos\left(\pm \frac{\pi t}{T_b}\right) \cos(2\pi f_c t) - \sqrt{\frac{2E_b}{T_b}} \sin\left(\pm \frac{\pi t}{T_b}\right) \sin(2\pi f_c t)$$

$$= \sqrt{\frac{2E_b}{T_b}} \cos\left(\frac{\pi t}{T_b}\right) \cos(2\pi f_c t) \mp \sqrt{\frac{2E_b}{T_b}} \sin\left(\frac{\pi t}{T_b}\right) \sin(2\pi f_c t)$$

In the above equation, the '+' sign corresponds to transmitting symbol 0 and the '−' sign corresponds to transmitting symbol 1.

Also, the inphase component is completly independent of the input binary wave. It equals $\sqrt{\frac{2E_b}{T_b}} \cos(\pi t/T_b)$ for all time t. The PSD consists of two delta functions and weighted by a factor $E_b/2T_b$ occuring at $f = \pm \frac{1}{2T_b}$

The Quadrature component is directly related to the input binary sequence. During the signaling interval $0 \le t \le T_b$, it equals $-g(t)$ when symbol is 1 and $+g(t)$ when symbol is 0, with $g(t)$ is a pulse shaping function.

$$g(t) = \sqrt{\frac{2E_b}{T_b}} \sin(\pi t/T) \quad , \quad 0 \le t \le T$$

Energy Spectral density of $g(t)$ is

$$\varphi_g(f) = \frac{8E_bT_b\cos^2(\pi T_b f)}{\pi^2(4T_b^2f^2-1)^2}$$

The total PSD is the sum of the PSD's of inphase and quadrature components as given by

$$S_{XX}(f) = \frac{E_b}{2T_b}\left[\delta\left(f-\frac{1}{2T_b}\right)+\delta\left(f+\frac{1}{2T_b}\right)\right]+\frac{8E_bT_b\cos^2(\pi T_b f)}{\pi^2(4T_b^2f^2-1)^2}$$

* The baseband PSD of a BFSK signal with continuous phase ultimately falls off as the inverse fourth power of frequency.

* A BFSK signal with continuous phase does not produce as much interference outside the signal band of interest as a corresponding FSK signal with discontinuous phase does.

* This property of CPBFSK will be utilized in MSK, which will be discussed further in detail.

In BFSK $f_1 = f_c + \frac{\Delta f}{2}$ and $f_2 = f_c - \frac{\Delta f}{2}$

$\Delta f \rightarrow$ frequency seperation

Modulation Index $= h = \frac{\Delta f}{R_b} = T_b\Delta f$

$\Delta f_{min} = \frac{1}{T_b}$ for non-coherent FSK detection

$\Delta f_{min} = \frac{1}{2T_b}$ for coherent FSK detection.

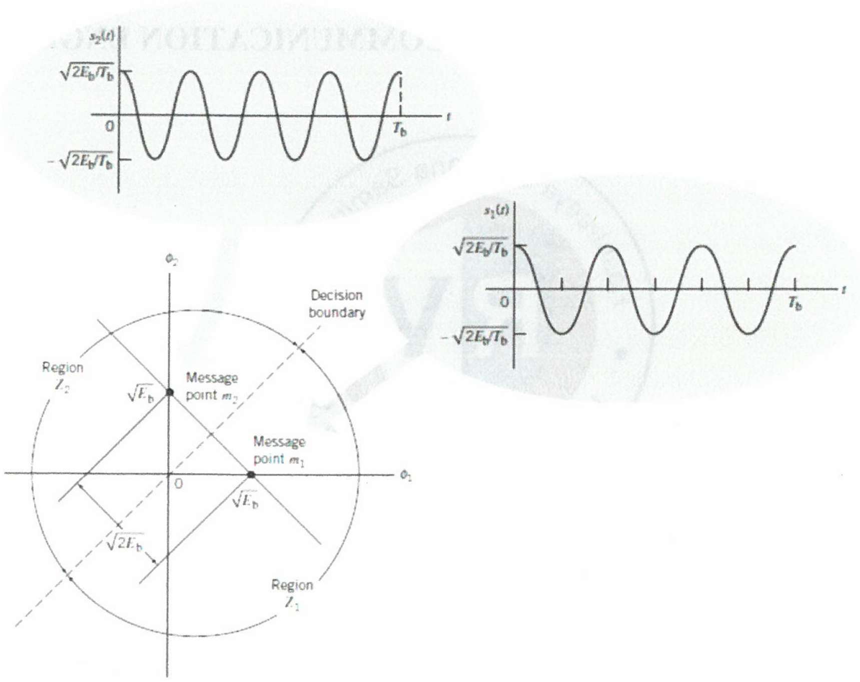

Figure 7.25 Signal-space diagram for binary FSK system. The diagram also includes example waveforms of the two modulated signals $s_1(t)$ and $s_2(t)$.

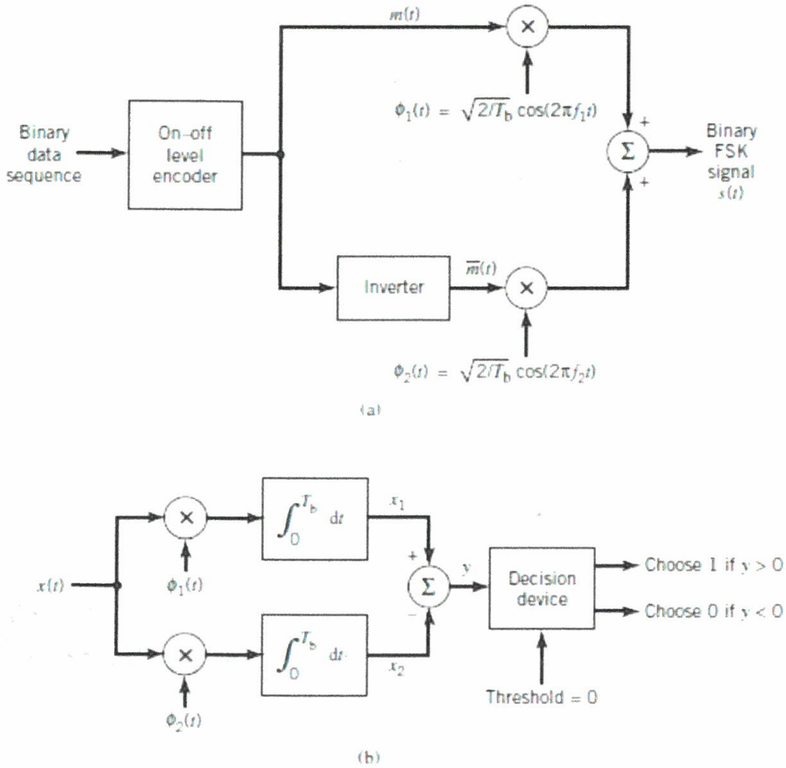

Figure 7.26 Block diagram for (a) binary FSK transmitter and (b) coherent binary FSK receiver.

Bandwidth Efficiency of M-ary FSK Signals

When the orthogonal signals of an M-ary FSK signal are detected coherently, the adjacent signals need only be separated from each other by a frequency difference $1/2T$ so as to maintain orthogonality. Hence, we may define the channel bandwidth required to transmit M-ary FSK signals as

$$B = \frac{M}{2T_s} \qquad T_s = kT_b = \frac{K}{R_b} \tag{7.207}$$

For multilevels with frequency assignments that make the frequency spacing uniform and equal to $1/2T_s$ the bandwidth B of (7.207) contains a large fraction of the signal power.

$T_s = kT_b = \log_2 M \cdot T_b$,

$$B = \frac{R_b M}{2 \log_2 M}$$

The bandwidth efficiency of M-ary signals is therefore

$$\rho = \frac{R_b}{B} = \frac{2 \log_2 M}{M}$$

We see that increasing the number of levels M tends to increase the bandwidth efficiency of M-ary PSK signals, but it also tends to decrease the bandwidth efficiency of M-ary FSK signals. In other words, M-ary PSK signals are spectrally efficient, whereas M-ary FSK signals are spectrally inefficient.

Table 7.5 Bandwidth efficiency of M-ary FSK signals

M	2	4	8	16	32	64
ρ (bits/(sHz))	1	1	0.75	0.5	0.3125	0.1875

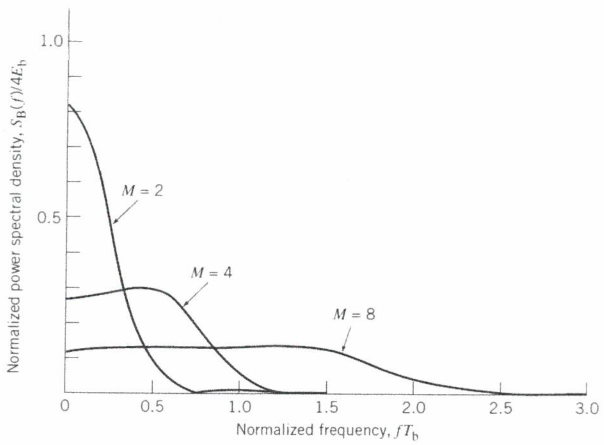

Figure 7.35 Power spectra of M-ary PSK signals for $M = 2, 4, 8$.

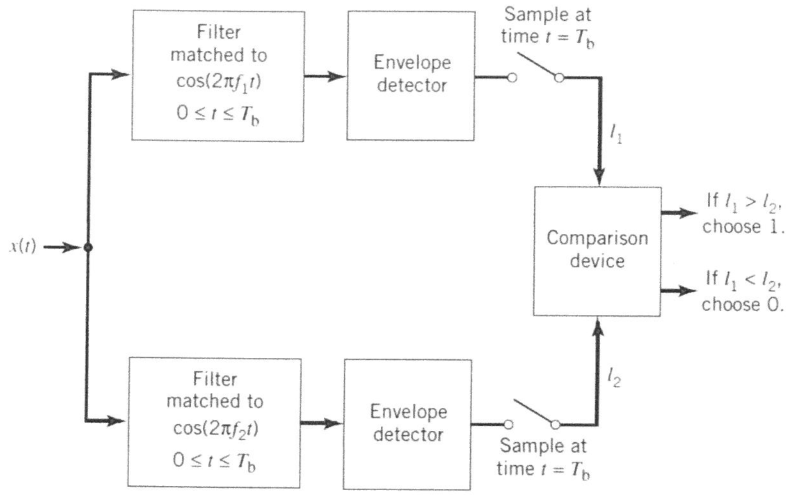

Figure 7.43 Noncoherent receiver for the detection of binary FSK signals.

Differential Phase-Shift Keying

we may view DPSK as the "noncoherent" version of binary PSK. The distinguishing feature of DPSK is that it eliminates the need for synchronizing the receiver to the transmitter by combining two basic operations at the transmitter:

- *differential encoding* of the input binary sequence and
- *PSK* of the encoded sequence, from which the name of this new binary signaling scheme follows.

Differential encoding starts with an arbitrary first bit, serving as the *reference bit*; to this end, symbol 1 is used as the reference bit. Generation of the differentially encoded sequence then proceeds in accordance with a two-part *encoding rule* as follows:

1. If the new bit at the transmitter input is 1, leave the differentially encoded symbol unchanged with respect to the current bit.
2. If, on the other hand, the input bit is 0, change the differentially encoded symbol with respect to the current bit.

The differentially encoded sequence, denoted by $\{d_k\}$, is used to shift the sinusoidal carrier phase by zero and 180^o, representing symbols 1 and 0, respectively. Thus, in terms of phase-shifts, the resulting DPSK signal follows the two-part rule:

1. To send symbol 1, the phase of the DPSK signal remains unchanged.
2. To send symbol 0, the phase of the DPSK signal is shifted by 180°.

Illustration of DPSK

Consider the input binary sequence, denoted $\{b_k\}$, to be 10010011, which is used to derive the generation of a DPSK signal. The differentially encoded process starts with the reference bit 1. Let $\{d_k\}$ denote the differentially encoded sequence starting in this manner and $\{d_{k-1}\}$ denote its delayed version by one bit. The complement of the modulo-2 sum of $\{b_k\}$ and $\{d_{k-1}\}$ defines the desired $\{d_k\}$, as illustrated in the top three lines of Table 7.6. In the last line of this table, binary symbols 1 and 0 are represented by phase-shifts of 1 and π radians.

Table 7.6 **Illustrating the generation of DPSK signal**

$\{b_k\}$		1	0	0	1	0	0	1	1
$\{d_{k-1}\}$		1	1	0	1	1	0	1	1
	reference								
Differentially encoded sequence $\{d_k\}$	1	1	0	1	1	0	1	1	1
Transmitted phase (radians)	0	0	π	0	0	π	0	0	0

Basically, the DPSK is also an example of noncoherent orthogonal modulation when its behavior is considered over successive two-bit intervals; that is, $0 \le t \le 2T_b$. To elaborate, let the transmitted DPSK signal be $\sqrt{2E_b/T_b}\cos(2\pi f_c t)$ for the first-bit interval $0 \le t \le T_b$, which corresponds to symbol 1. Suppose, then, the input symbol for the second-bit interval $T_b \le t \le 2T_b$ is also symbol 1. According to part 1 of the DPSK encoding rule, the carrier phase remains unchanged, thereby yielding the DPSK signal

$$s_1(t) = \begin{cases} \sqrt{\dfrac{2E_b}{T_b}}\cos(2\pi f_c t), & \text{symbol 1 for} \quad 0 \le t \le T_b \\ \sqrt{\dfrac{2E_b}{T_b}}\cos(2\pi f_c t), & \text{symbol 0 for} \quad T_b \le t \le 2T_b \end{cases} \tag{7.246}$$

Suppose, next, the signaling over the two-bit interval changes such that the symbol at the transmitter input for the second-bit interval $T_b \le t \le 2T_b$ is 0. Then, according to part 2 of the DPSK encoding rule, the carrier phase is shifted by π radians (i.e., 180°), thereby yielding the new DPSK signal

$$s_2(t) = \begin{cases} \sqrt{\dfrac{2E_b}{T_b}}\cos(2\pi f_c t), & \text{symbol 1 for} \quad 0 \le t \le T_b \\ \sqrt{\dfrac{2E_b}{T_b}}\cos(2\pi f_c t + \pi), & \text{symbol 1 for} \quad T_b \le t \le 2T_b \end{cases} \tag{7.247}$$

We now readily see from (7.246) and (7.247) that $s_1(t)$ and $s_2(t)$ are indeed orthogonal over the two-bit interval $0 \le t \le 2T_b$, which confirms that DPSK is indeed a special form of noncoherent orthogonal modulation with one difference compared with the case of binary FSK: for DPSK, we have $T = 2T_b$ and $E = 2E_b$. Hence, using (7.227), we find that the *BER for DPSK* is given by

$$P_e = \frac{1}{2}\exp\left(-\frac{E_b}{N_0}\right) \tag{7.248}$$

According to this formula, DPSK provides a gain of 3 dB over binary FSK using noncoherent detection for the same E_b/N_0.

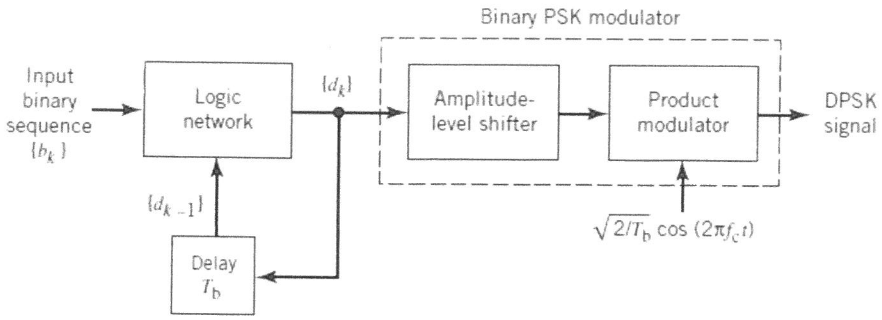

Figure 7.44 Block diagram of a DPSK transmitter.

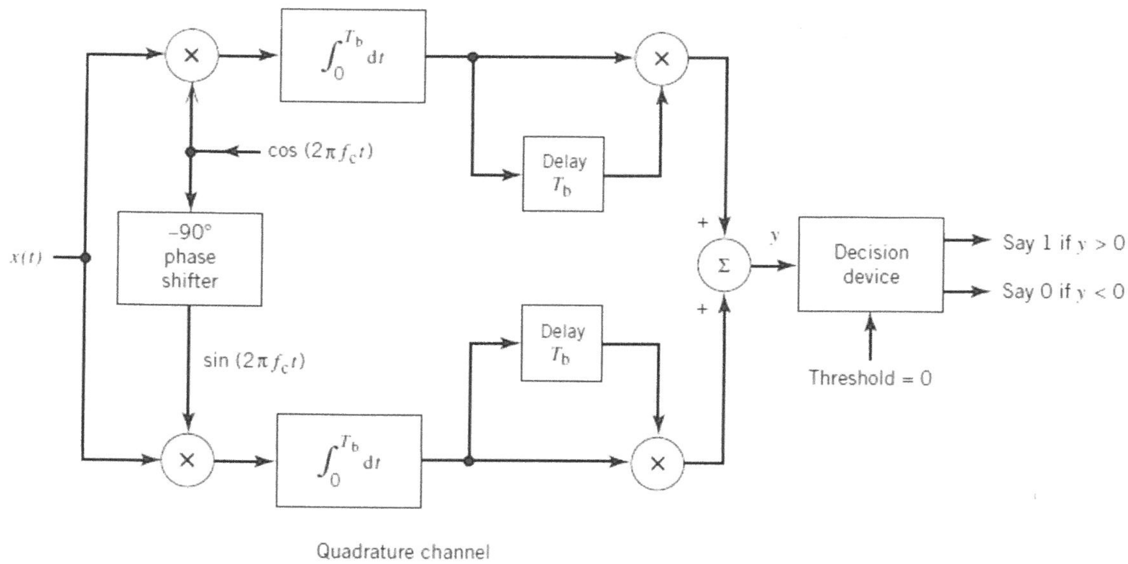

Figure 7.46 Block diagram of a DPSK receiver.